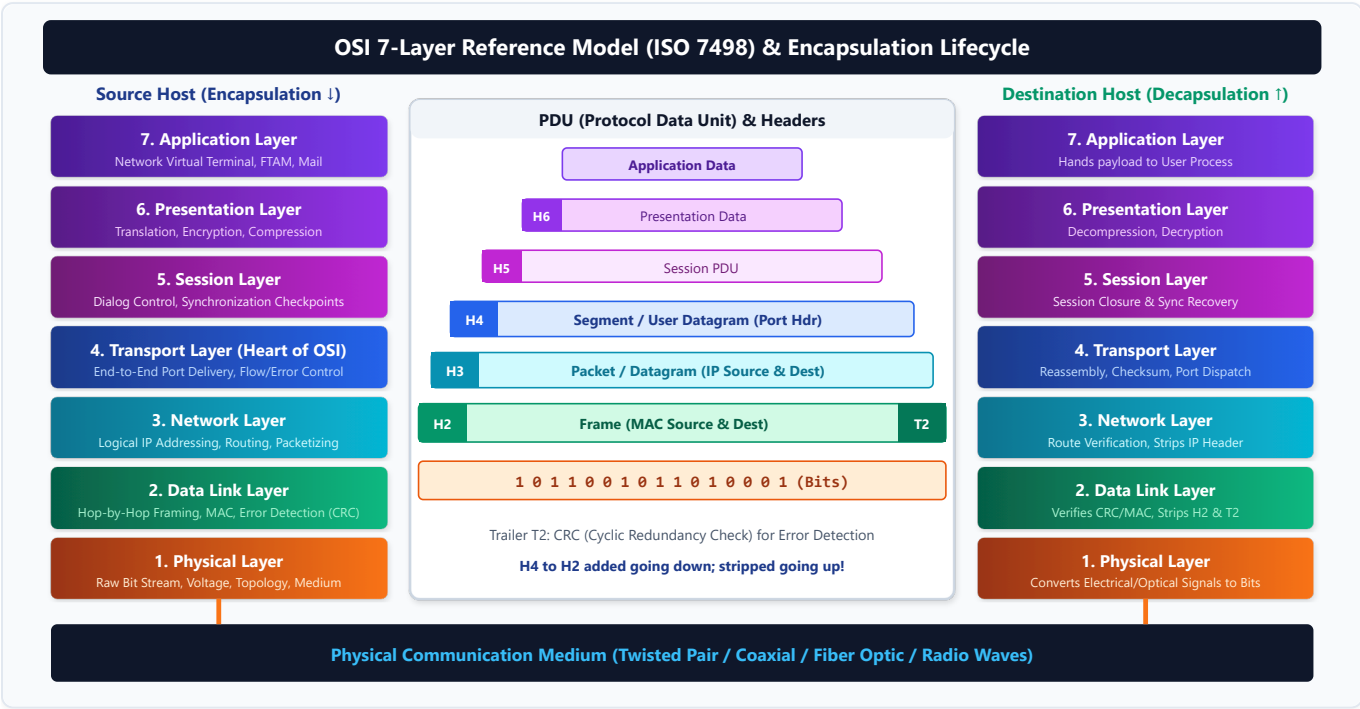


Module 05: OSI Reference Model Architecture & 7 Layers

AKTU BCS603 Core Syllabus: ISO 7498 specification, functions of all 7 layers, PDU mapping, encapsulation headers, and trailer anatomy.



1. Master 7-Layer Summary Matrix

Layer # & Name	PDU	Primary Responsibility	Key Addressing / Identifiers	Associated Protocols / Hardware
7. Application	Data / Message	User UI & Network Service Access	URI / URL, Email IDs	HTTP, FTP, SMTP, DNS, SSH, SNMP
6. Presentation	Data	Data Translation, Encryption, Compression	MIME types, Character sets	SSL/TLS, ASCII, JPEG, MPEG, gzip
5. Session	Data	Dialog establishment, Checkpoints & Sync	Session IDs, Token controls	NetBIOS, PPTP, RPC, SIP
4. Transport	Segment / Datagram	End-to-End Process Delivery, Flow & Error	Port Numbers (16-bit: 0 - 65535)	TCP, UDP, SCTP
3. Network	Packet	Source-to-Destination Host Routing	Logical IP Address (32/128-bit)	IP, ICMP, OSPF, BGP <i>Routers</i>
2. Data Link	Frame	Hop-to-Hop Reliable Frame Delivery	Physical MAC Address (48-bit)	Ethernet 802.3, Wi-Fi 802.11 <i>Switches</i>
1. Physical	Bits	Transmission of raw binary stream	Voltages, RF carrier frequencies	EIA-232, RJ-45, Cat6, Fiber <i>Hubs</i>

2. Network Layer vs Transport Layer Delivery (Exam Favorite)

- Network Layer:** Sirf packet ko Source Host computer se Destination Host computer tak pahunchata hai (Host-to-Host delivery). Isse yeh nahi pata hota ki destination computer par kaunsa application software packet ka intezaar kar raha hai.

- **Transport Layer:** Packet ke andar ke Port Number ko inspect karke data ko destination computer ke exact running software process (e.g., Chrome tab vs Spotify) tak pahunchata hai (Process-to-Process delivery).

AKTU Exam Alert: "Describe all layers of the OSI model with a well-labeled diagram" 10-mark question me har layer ka PDU, 2-3 main functions, aur trailer \$T_2\$ ka role zaroor mention karein!